GHAYDA NA'IM JA'AFREH

Data Science & Artificial Intelligence Graduate | Python | Machine Learning | AI & Software Systems

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PROFESSIONAL SUMMARY

Data Science and Artificial Intelligence graduate with an Excellent GPA of 93.04%, ranked second in the graduating cohort. Hands-on experience across Python programming, machine learning, deep learning, computer vision, data analysis, backend integration, REST APIs, model evaluation, debugging, and technical documentation. Built applied projects spanning healthcare AI, multimodal authentication, robotics integration, autonomous gameplay, image segmentation, interactive ML applications, and reproducible experimentation. Strong analytical and problem-solving skills with the ability to translate technical requirements into practical solutions.

TECHNICAL SKILLS

Programming & Software: Python, SQL, C++, PHP, HTML, CSS, JavaScript, modular programming, debugging, input validation, error handling, Git, GitHub

AI & Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, classification, regression, CNNs, transfer learning, segmentation, explainable AI, model evaluation

Data & Analytics: Pandas, NumPy, data cleaning, preprocessing, feature engineering, visualization, comparative experimentation

Backend & Integration: FastAPI, REST APIs, application integration, logging, dependency management, virtual environments, interface validation

Computer Vision & Tools: OpenCV, image processing, U-Net, OCR, Gradio, Pygame, Jupyter Notebook, Azure-hosted GPU environment, Excel, Tableau

Databases: SQLite, MySQL, SQL workflows

EDUCATION

Bachelor's Degree in Data Science and Artificial Intelligence | Mutah University | 2022-2026

GPA: 93.04% | Grade: Excellent | Ranked 2nd in the Data Science and Artificial Intelligence graduating cohort

SELECTED PROJECTS

StrokeLens - Secure & Explainable Healthcare AI Graduation Project

Graduation Project | Sep 2025-Jun 2026 | Core Project Owner and Lead Technical Contributor

- Led an end-to-end healthcare AI project using structured clinical data, multi-stage stroke-risk modeling, explainability, secure reporting, and integrated application workflows.
- Contributed across data preparation, model experimentation and evaluation, threshold analysis, explainability, system integration, testing, technical documentation, and Azure-hosted GPU experimentation.

Technologies: Python, machine learning, explainable AI, model evaluation, secure systems, Azure GPU

Privacy-Safe Multimodal Authentication System

Team Project | Main Voice-Model and Integration Contributor

- Led work on the voice-verification component and contributed to integration, verification logic, API-related workflow, testing, logging, and privacy-aware data handling.
- Integrated voice verification with a Python/FastAPI multimodal workflow that also incorporated face verification and optional offline speech recognition; resolved runtime, dependency, and input-format issues through systematic testing.

Technologies: Python, FastAPI, PyTorch, SpeechBrain, OpenCV, InsightFace, Vosk, REST APIs, authentication, logging

Tetris AI Lab - End-to-End AI Game Application

Individual Project | Public GitHub Release

- Developed and published a Python/Pygame application integrating a CNN policy with inference, top-5 one-piece lookahead, safety controls, replay functionality, and human/AI gameplay modes.
- Built a reproducible training-data pipeline with automated cleaning, video-level train/validation/test splitting, simulator-generated recovery states, simulator-verified mirror augmentation, experiment tracking, testing, and release documentation.

Technologies: Python, PyTorch, CNN, Pygame, data pipelines, testing, reproducible experimentation

Dr. NAO - AI-Assisted Dermatology Support System

Team Project | Main AI/Model and Robot-Integration Contributor

- Integrated AI model outputs with a NAO robot interaction workflow for a healthcare prototype presented in a live academic demonstration.
- Supported model and robot integration, interaction logic, testing, troubleshooting, demonstration preparation, and technical explanation.

Technologies: Python, computer vision, AI integration, NAO robotics, testing, troubleshooting

Medical Image Segmentation & Computer Vision Portfolio

Kaggle and Academic Projects

- Built teeth instance segmentation for dental X-rays, achieving Mask mAP50 of 0.9947 and Mask mAP50-95 of 0.6301.
- Implemented a custom PyTorch U-Net workflow for dental-caries segmentation, achieving 0.9010 test Dice; additional work includes Intel transfer learning, MNIST CNN experiments, OCR with HOG/KNN, and image classification.

Technologies: Python, PyTorch, OpenCV, U-Net, transfer learning, OCR, segmentation

ML Decision Surfaces Lab - Interactive Machine Learning Playground

Team Project | Main Algorithm Developer and Interface Designer

- Built an interactive Gradio application for comparing classification and regression models using decision boundaries, ROC-AUC, confusion matrices, learning curves, and controlled noise/outlier experiments.

Technologies: Python, scikit-learn, Gradio, Matplotlib, model evaluation

EXPERIENCE

Data Science & Programming Trainee | The Hope International Company | Online Training | Jun 20, 2025-Sep 10, 2025

- Completed structured practical training in Python programming, data analysis, preprocessing, visualization, model implementation, debugging, teamwork, and technical documentation.
- Applied analytical and programming skills to structured datasets and small end-to-end AI workflows.

AI & Machine Learning Trainee | The Hope International Company | Online Training | Nov 15, 2023-Mar 15, 2024

- Built and evaluated Python-based machine-learning exercises covering data preparation, feature selection, classification, regression, decision trees, KNN, Naive Bayes, and model evaluation.
- Developed end-to-end understanding of the machine-learning lifecycle from preparation through evaluation and reporting.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft | Earned May 27, 2026 | Credential ID: C5834AFC20D9A41C | Certification No.: B010AT-7276E5
- Data Science Essentials with Python | Cisco Networking Academy | May 29, 2026 | Certificate ID: 960f2b70-a252-42a1-a73d-13e7ca5c7c51
- Python Essentials 1 | Cisco Networking Academy | May 30, 2026 | Certificate ID: 01720f08-f6be-48a8-b730-d4da54339d1f
- Deep Learning - 60 Hours | The Hope International Company | Sep 15, 2025 | Certificate ID: 1758145099980
- Python AI (Machine Learning) - 60 Hours | The Hope International Company | Nov 2023-Mar 2024
- Ethical Hacker | Cisco Networking Academy | May 29, 2026 | Certificate ID: 34898e2b-05cf-41a1-bcc0-c8d3f49090d1
- Cyber Security Associate - Certificate of Completion | Green Circle for Software Solutions | 2026
- CCNA Course Certificate | Pioneers Academy | Dec 28, 2023 | Certificate No.: 18437657-24
- Security+ Course Certificate - 30 Hours | Pioneers Academy | Oct 5, 2023 | Certificate No.: 18424325-2560
- Human Research: Data or Specimens Only Research - Basic Course | CITI Program, MIT Affiliates | Jul 13, 2025-Jul 13, 2028 | Record ID: 70859362
- HTML Essentials | Cisco Networking Academy | May 30, 2026 | Certificate ID: 1ca22c8d-bbae-4328-8cf8-07eacac17e5e
- International Computer Driving License (ICDL) - 36 Hours | Arab International Academy | Feb 5, 2024 | CRT.No: 201548
- English Communication Skills Training - 20 Hours | Mutah University / King Abdullah II Fund for Development | Apr 6-May 13, 2024
- English Conversation - 12 Levels, 90 Hours | Arab International Academy | Feb 1-May 15, 2023 | ID: 200166016 | RN: 117840

LANGUAGES & CAREER FOCUS

Languages: Arabic (Native), English (Intermediate)
Career Focus: Artificial Intelligence, Machine Learning, Data Science, Computer Vision, Python Development, Backend Integration, Analytics, AI Research, and Software Engineering.